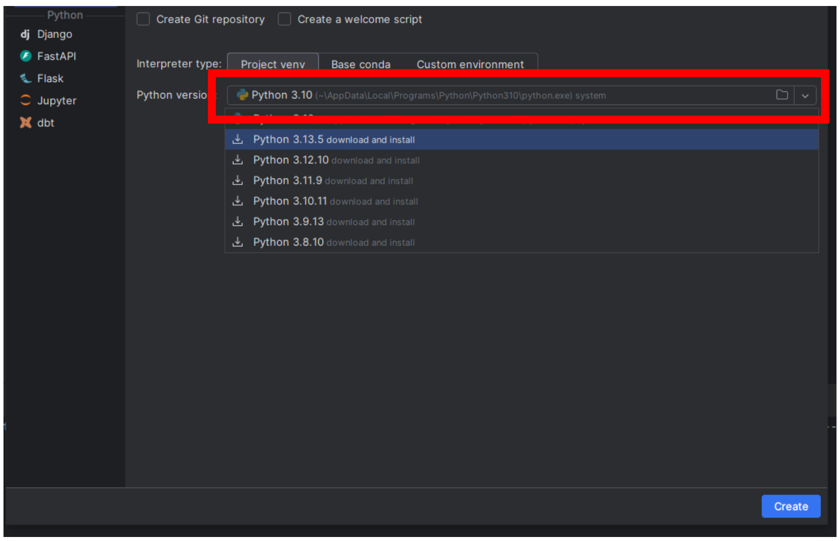
Once you have downloaded the BSD-QBio repository from GitHub and have installed PyCharm on your computer, please launch PyCharm, then follow these steps.

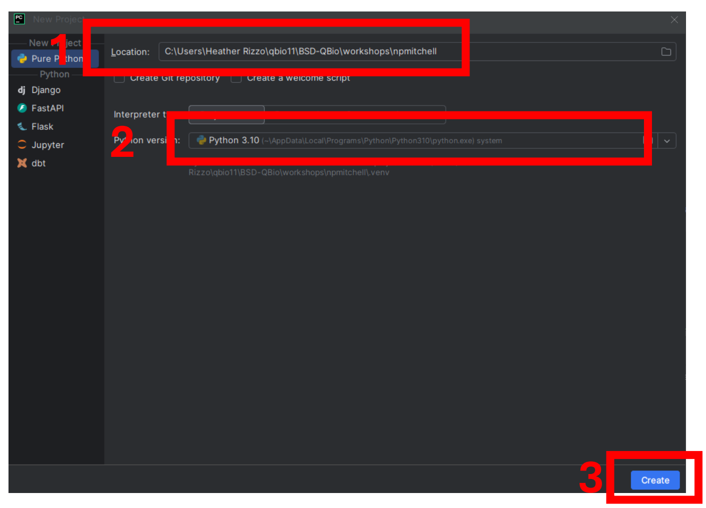
Step 1

Go to File > New Project...

For the workshop, we will need Python 3.10. Under “Python version”, select 3.10. If necessary, it will automatically download and install this version of Python. (If this does not work, please see “Troubleshooting” at the end.)



1. The location of your new project should be the same as the folder containing the materials for the Biological Shape Analysis workshop (.../BSD-QBio/workshops/npmitchell). Note that the name of the folder containing the workshop materials (“npmitchell”) will also be the name of your project.
2. Check that python version 3.10 is being used.
3. Click create!



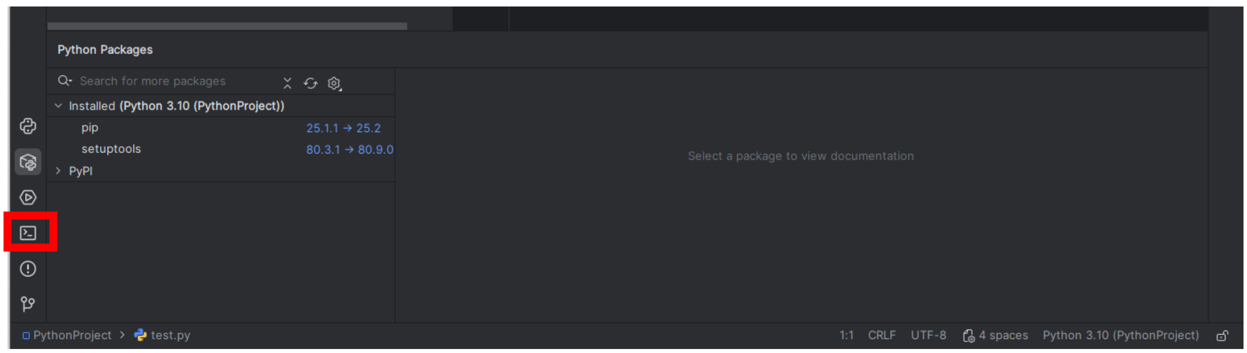
Step 2

Install the Python packages you will need for the workshop.

Open the Terminal (red box in image below) in PyCharm, and install the packages by typing

pip install numpy scipy pandas matplotlib pyvista open3d

then hit the enter or return key.



To check that the packages have been installed successfully, go the Python Console (it is one of the icons above the Terminal icon) and run the following lines of Python code:

import numpy
import pandas
import matplotlib
import pyvista
import scipy
import open3d

If you do not get an error when running these lines of code, then you have installed the packages correctly.

Step 3

Finally, the ZIP file(s) you downloaded from Box should be unzipped and moved to the "npmitchell" folder so that the "wildtype" and "bynGAL4_UASMyo1C" folders are inside the "npmitchell" folder. (You should also see the other workshop files in that folder such as npmitchell_workshop.Rmd and mesh.py.) In short, all the files needed for the Biological Shape Analysis workshop — the Python code and the data files — should be in the same folder.

Troubleshooting

We will need to have Python version 3.10 installed (do not install the latest version). If it does not install automatically when you create a new project, try this link (scroll down on the webpage to access the Python installers).

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		729e36388ae9a832b01cf9138921b383	23.8 MB	SIG
XZ compressed source tarball	Source release		3e7035d272680f80e3ce4e8eb492d580	17.9 MB	SIG
macOS 64-bit universal2 installer	macOS	for macOS 10.9 and later (updated for macOS 12 Monterey)	8575cc983035ea2f0414e25ce0289ab8	37.9 MB	SIG
Windows installer (64-bit)	Windows	Recommended	c3917c08a7fe85db7203da6dcaa99a70	27.0 MB	SIG
Windows installer (32-bit)	Windows		133aa48145032e341ad2a000cd3bff50	25.9 MB	SIG
Windows help file	Windows		9d7b80c1c23cfb2cecd63ac4fac9766e	9.1 MB	SIG
Windows embeddable package (64-bit)	Windows		340408540eeff359d5eaf93139ab90fd	8.1 MB	SIG
Windows embeddable package (32-bit)	Windows		dc9d1abc644dd78f5e48edae38c7bc6b	7.2 MB	SIG